FDIR Copilot

An AI flight controller that finds the root cause of small-satellite anomalies, and learns from every incident.

93%

root-cause accuracy vs 32% limit-check

50%

of faults never alarm on a classic console

70 s

median detection vs 450 s

Learning from memory: **76% → 100%** accuracy as closed incidents accumulate.

WHO IT'S FOR

NewSpace start-ups and university CubeSat teams: 2 to 5 operators flying many spacecraft, with no 24/7 flight-controller bench.

THE PROBLEM

Consoles alarm on limit violations: symptoms (battery temperature), not causes (a degrading solar array). Half of faults never cross a limit at all.

WHY NOW

Thousands of small-sat launches a year, flat ops headcount, and agents that can reason over telemetry plus a causal model of the vehicle.

ROUTE TO FIRST USERS

Free tier for CubeSat teams (YAMCS / OpenMCT / CCSDS importers) → paid pilots with 3 NewSpace operators, per spacecraft per month → SmallSat Conference, LibreCube, launch brokers.

HOW THE AGENT WORKS

Telemetry downlink → Aiven PostgreSQL

① Detect: orbit-aware residuals (eclipse ephemeris)

② Isolate: backtrack a system-dynamics causal graph

③ Recall: kNN over incident memory in Aiven PostgreSQL

④ Fuse evidence ⑤ Recommend recovery + briefing

Operator closes incident → memory → next pass is smarter

POINT OF DIFFERENCE

Built on aerospace systems engineering (FDIR, FMEA, system dynamics). It explains why, with a traceable causal chain, and runs on Aiven Runtime.

THE PROBLEM

Consoles alarm on symptoms. Operators need causes.

Small-sat ops teams lose hours per anomaly on manual telemetry forensics.

A DEGRADING SOLAR ARRAY, AS THE CONSOLE SEES IT

sa_current ↓	+160 s	root cause, masked in eclipse
bus_voltage ↓	+450 s	in limits
batt_soc ↓	+800 s	in limits
batt_voltage ↓	+2990 s	in limits
batt_temp ↑	+4250 s	first thing an operator notices

49.7%

of injected faults never cross a single hard limit. A limit-check console stays silent while the spacecraft drifts toward safe mode.

32%

root-cause accuracy when you blame the subsystem of the first alarm, which is what a limit-check console effectively does.

THE SOLUTION

A 5-step agent that reasons like a flight controller

Deterministic, traceable core. The LLM only writes the human briefing.

01

Detect

Orbit-aware residuals, eclipse-conditioned baselines, persistence filter

02

Isolate

Backtrack the spacecraft's causal graph: root, sign, precedence, coverage

03

Recall

kNN over closed incidents stored in Aiven PostgreSQL

04

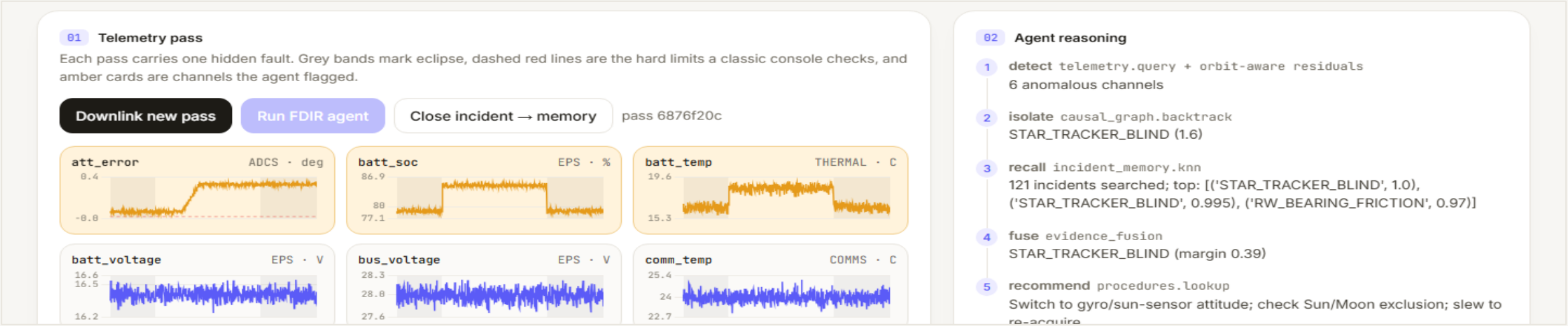
Fuse

Combine causal and memory evidence; confidence from the margin

05

Recommend

Recovery procedure and operator briefing; full trace logged



DATA AT THE CORE

Aiven PostgreSQL: the agent's memory

Without the stored telemetry, ephemeris grounding and incident memory, the agent cannot diagnose anything.

Downlink
13 channels · 10 s
9,360 rows / pass

→

`telemetry`
Aiven PostgreSQL
passes · telemetry

→

FDIR agent
detect → isolate →
recall → fuse → recommend

→

`agent_runs`
Aiven PostgreSQL
audit trail + latency

→

`incidents`
Aiven PostgreSQL
long-term memory

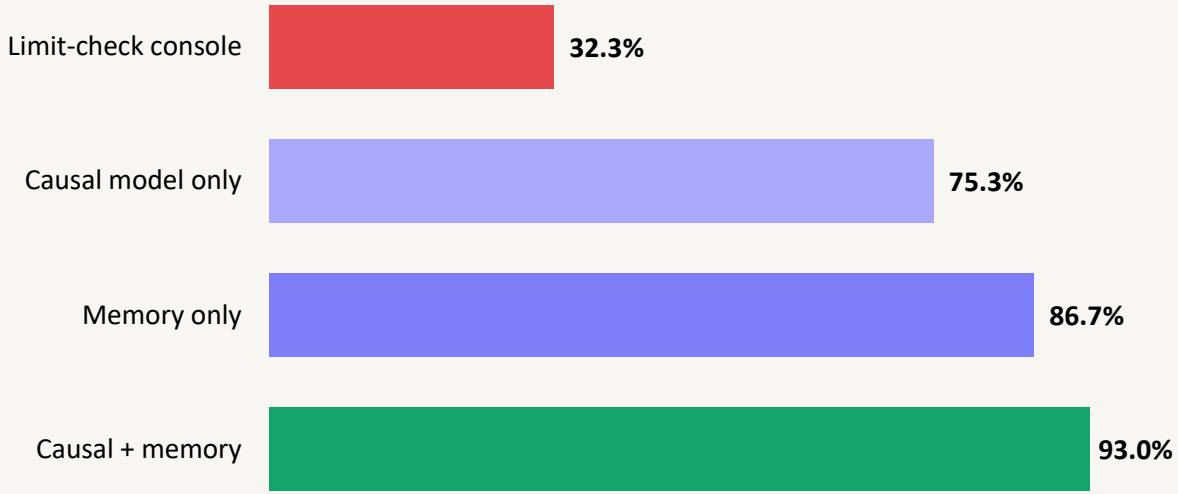
↺ **Operator closes the incident → confirmed root cause written to incidents → the next pass is diagnosed with more memory**

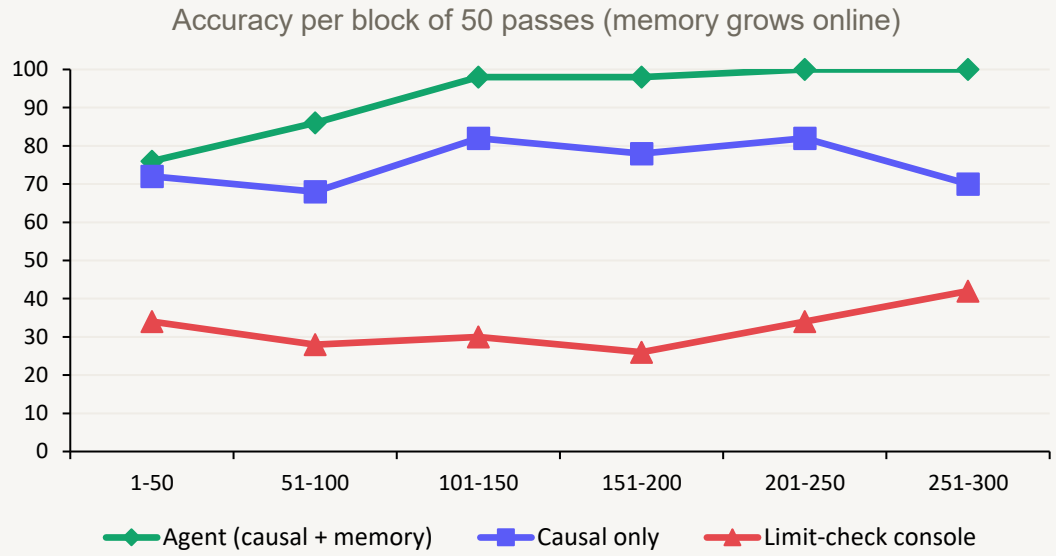
Data	Where it lives	Role
Telemetry passes	Aiven PostgreSQL · passes, telemetry	Raw evidence queried on every run
Orbit ephemeris	Derived per pass (eclipse / sunlit)	Grounds baselines; 0 A solar current in eclipse is nominal
Causal model	Gains + transport delays (system dynamics)	Separates causes from symptoms
Incident memory	Aiven PostgreSQL · incidents	kNN recall; grows with every closed incident
Agent run log	Aiven PostgreSQL · agent_runs	Audit trail for flight-safety review

EVIDENCE IT WORKS

Root-cause accuracy: 93% vs 32%

300 fault-injected passes · 6 fault classes · random onset, severity and orbit phase · memory starts empty





49.7%

faults the console never alarms; the agent flags them

70 s

median detection delay vs 450 s for the console

4 ms

median compute per diagnosis

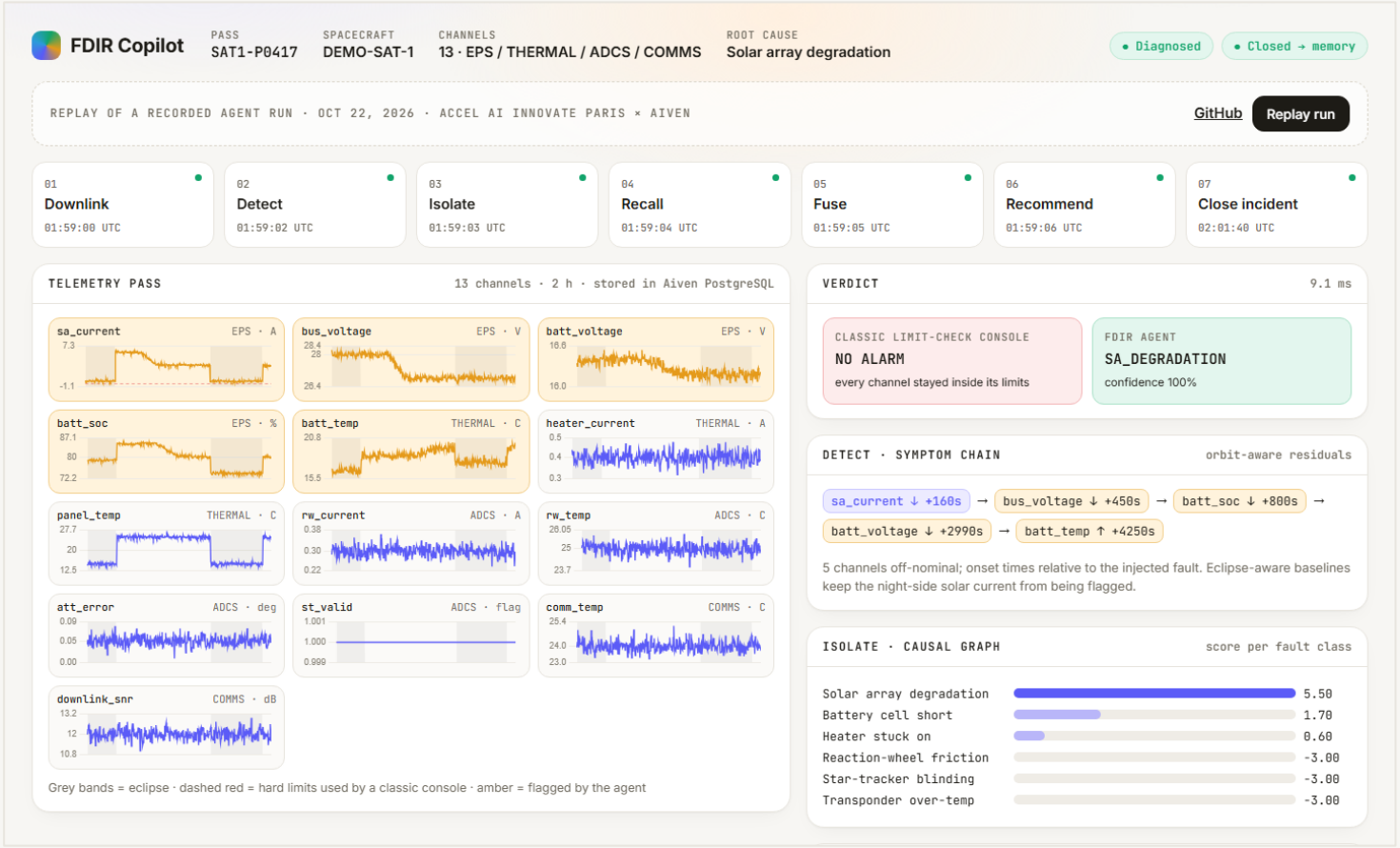
76→100%

agent accuracy as incident memory fills

PRODUCT DEMO

Live app on Aiven Runtime, plus a public replay

The console stays silent; the agent names the root cause, cites its memory and gives the recovery procedure.



REPLAY (GITHUB PAGES)

parasadlakha.github.io/fdir-copilot

SOURCE CODE

github.com/ParasAdlakha/fdir-copilot

DATA LAYER

Aiven for PostgreSQL

DEPLOYMENT

Aiven Runtime · Dockerfile · port 8080

From CubeSat teams to NewSpace fleets

ROUTE TO FIRST USERS

- Free tier for university CubeSat teams
- Open-source importers: YAMCS, OpenMCT, CCSDS
- Paid pilots with 3 seed-stage NewSpace operators
- Priced per spacecraft per month
- SmallSat Conference, LibreCube, launch brokers

ROADMAP

- Validate on NASA SMAP / MSL telemetry
- Learn causal gains and delays from fleet data
- Streaming via Aiven for Apache Kafka
- Long-horizon trends in Aiven for ClickHouse
- Fleet memory: satellite A pre-warns satellite B

TEAM

- Paras Adlakha: aerospace, systems engineering, system dynamics
- Harshita Adlakha: co-builder
- Built in one day at the hackathon

BUILT AT ACCEL AI INNOVATE PARIS · FUTURE FOUNDERS · 22 OCTOBER 2026

Partners: **Aiven · Lovable · Hackathon Hub · Iterate**